

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is that when the security team used a packet sniffer to capture the data in transit to and from the web server, we noticed that a lot of TCP SYN requests were coming from an unfamiliar IP address.

The web server appears to be overwhelmed by the volume of incoming traffic and is losing its ability to respond to the abnormally large number of SYN requests. We suspect the server is under attack by a malicious actor.

The actions taken to mitigate the risk was to turn the server offline so that the machine can recover and return to the normal operating status. From the logs results we came up with the idea that we encounter a Denial of Service attack standing the facts that only one device is being used to flood the server with SYN packets.

Section 2: Explain how the attack is causing the website to malfunction

We can see in the record log no. 47.51 that when the authorized source IP address is reaching to the destination IP address to form a three way handshake by sending the packets via a TCP protocol everything is occurring normally because the protocol changes to a HTTP meaning that was successfully established. We can see in the log that a SYN request it's sent to start a connection with the web server, the web server replies with SYN-ACK receiving the request and then the client sends a final ACK packet establishing a successful connection. An employee visitor with the IP address of 198.51.100.14 successfully completes a SYN/ACK connection handshake with the webserver (log item nos. 55, 56, 58). Then, the employee's browser requests the sales.html webpage with the GET command and the web server responds (log item no. 60 and 62).

Therefore we can see that the threat actor took advantage of the TCP protocol and it's trying to flood the server with multiple SYN requests affecting the first part of the TCP/IP handshake, so the attacker is targeting the bandwidth and he is trying to slow down the traffic. Initially, the attacker's SYN request is answered normally by the web server (log items 52-54). However, the attacker keeps sending more SYN requests, which is abnormal.

The two types of errors in the logs include an HTTP/1.1 504 Gateway Time-out (text/html) error message. This message was generated by the gateway server that was waiting for a response from the web server. The web server took too long to respond, so the gateway server sent a timeout error message to the requesting browser and an [RST, ACK] packet, which would be sent to the requesting visitor if the [SYN, ACK] packet is not received by the web server. RST stands for reset, acknowledge. That's why we received a timeout error message in the browser and the connection attempt was dropped. Potential ways to prevent SYN attacks include enabling SYN cookies, configuring firewalls and IDS/IPS systems, applying rate limit, reducing SYN timeouts values, blocking spoofed IP addresses, using DDoS protection services and monitoring traffic for abnormal SYN activity.